

CS4430 – Network Security Assignment 6: Network Telescope Analysis

Zubair Al-Zubi (5342007)

June 13, 2025

This report shows the results of analyzing network traffic from a telescope sensor. The goal is to find scanning behavior, top scanner IPs, and protocol use. The analysis was done using Python scripts on the provided PCAP files.

1 Top Scanners

Table 1 shows the ten most active scanning sources by packet volume. These IPs generated over 9% of total traffic, with 185.242.226.50 contributing 188,000+ packets, indicating automated scanning tools or botnet activity.

IP Address	Packets	Share
185.242.226.50	188,293	2.32%
185.242.226.46	105,263	1.30%
45.146.224.132	74,096	0.91%
10.14.176.3	72,407	0.89%
10.241.128.246	72,326	0.89%
80.75.212.37	59,683	0.74%
183.136.225.42	56,189	0.69%
212.70.149.138	54,256	0.67%
118.123.105.92	51,992	0.64%
218.92.0.99	49,420	0.61%

Table 1: Top 10 scanner IP addresses by packet count

2 Top Target Ports

Table 2 lists the top destination ports targeted by TCP and UDP scanners. Most TCP activity focused on ports commonly associated with remote access or web services (e.g., 22, 80, 443), while UDP scanning targeted DNS, SIP, and network service discovery. These results reflect broad automated probing for known services.

Port	Packets
22	174,938
3389	101,839
80	90,764
8088	82,746
8080	80,344
8728	76,159
5555	69,330
443	53,354
8122	51,620
12103	50,686

(a) TCP Scans

Port	Packets
514	144,733
53	91,129
5060	63,392
1194	40,973
32410	27,957
123	23,558
55001	13,926
1900	13,211
6881	13,009
5353	11,856

(b) UDP Scans

Table 2: Top 10 destination ports for TCP and UDP scans

3 Protocol Breakdown

Figure 1 shows the protocol breakdown of the captured traffic. TCP dominates with 7,404,342 packets (91.4%) versus UDP’s 697,351 packets (8.6%). This reflects typical scanning patterns targeting web and remote access services.

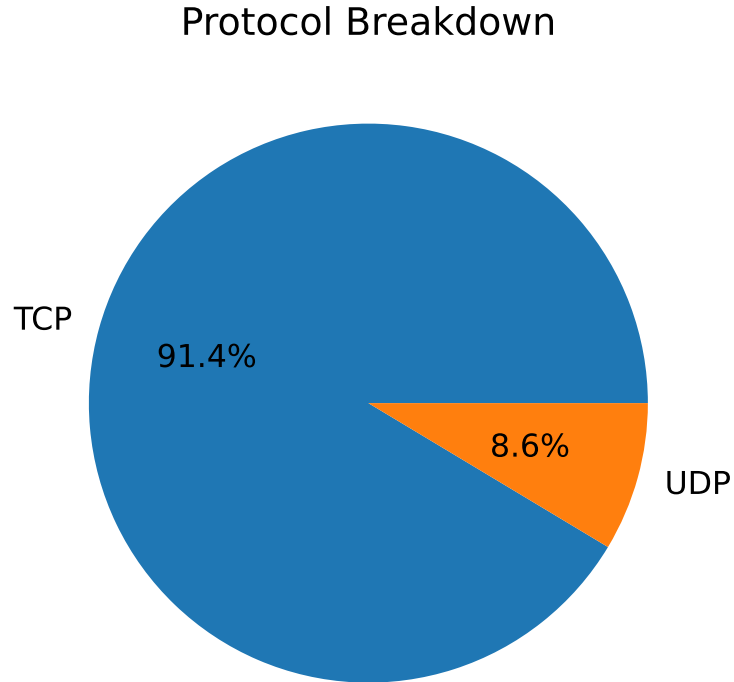


Figure 1: Protocol breakdown showing TCP vs UDP packet counts in the dataset

4 Heavy Hitter Scanner

Table 3 shows the scanner with the highest number of packets. IP address 185.242.226.50 sent over 188,000 TCP SYN and UDP packets, targeting 49,853 unique IP addresses and 4 different ports. This indicates a broad scanning strategy, likely automated. The IP geolocates to the Netherlands (country code NL).

Field	Value
Scanner IP	185.242.226.50
Number of packets	188,293
Number of destination IPs	49,853
Number of ports	4
Country	NL

Table 3: Profile of the heavy-hitter scanner

5 Visualization

5.1 Time Series

Figure 2 shows the number of scanning packets observed over time, grouped into 5-minute bins. The traffic starts at a peak of nearly 1.5 million packets and gradually decreases before showing a slight rise again near the end.

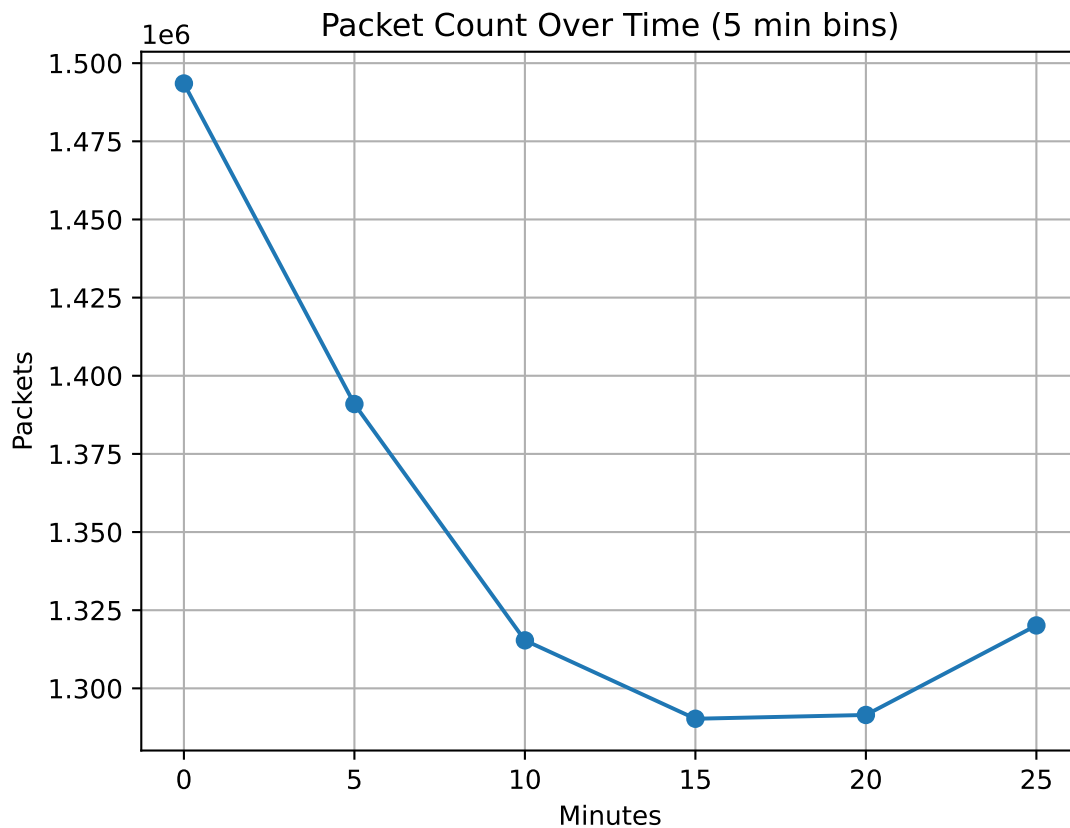


Figure 2: Packet count over time in 5-minute intervals

5.2 Port Distribution

Figure 3 shows the top 10 most scanned TCP and UDP destination ports. TCP scans target services such as SSH (22), RDP (3389), and HTTP(S), while UDP scans focus on DNS (53), syslog (514), SIP (5060), and NTP (123).

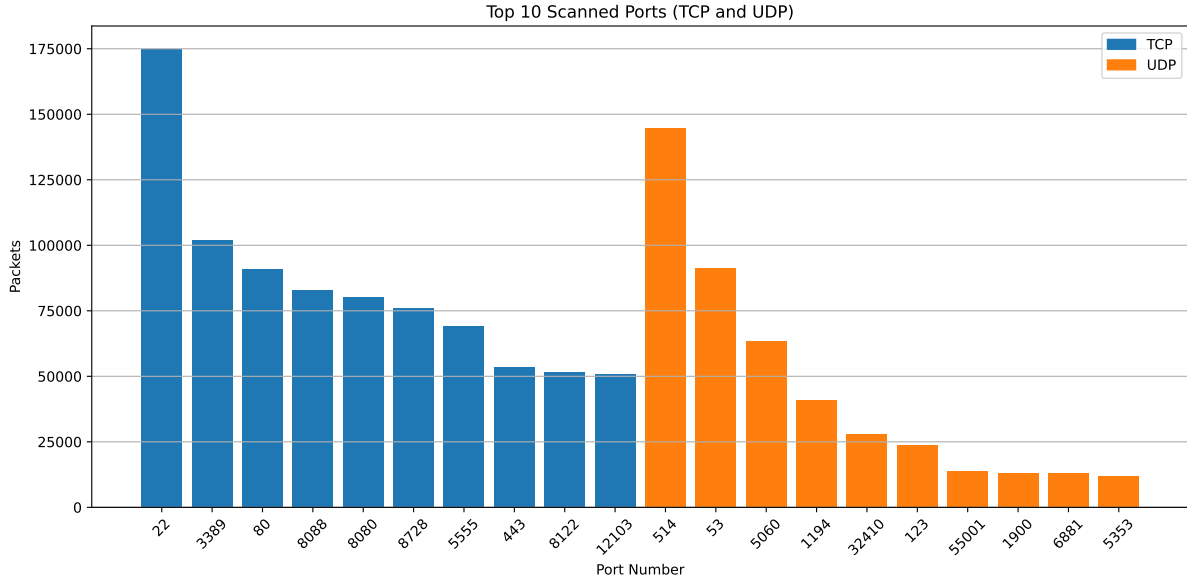


Figure 3: Top 10 scanned TCP and UDP ports

5.3 Scanner Scope

Figure 4 shows the number of unique destination IP addresses contacted by the top 10 scanner IPs. The most aggressive scanner contacted nearly 50,000 IPs.

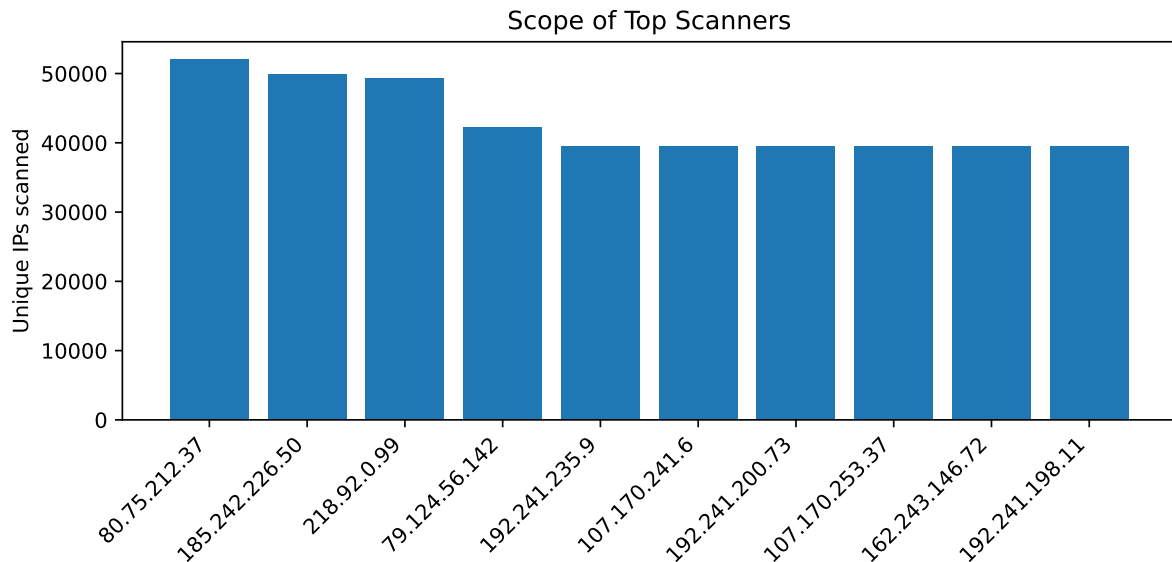


Figure 4: Number of unique IPs scanned by the top 10 scanners

6 Interpretation and Reasoning

6.1 Pattern Recognition

The overall scanning activity appears steady and persistent rather than bursty, as seen in [Figure 2](#), which shows relatively consistent traffic across the capture period. This suggests automated scanning tools operating continuously. In terms of target selection, [Figure 3](#) reveals a strong preference for a small set of well-known ports (e.g., 22, 3389, 80 for TCP and 514, 53 for UDP), which indicates service-specific reconnaissance rather than broad-spectrum port sweeps. This pattern aligns with mass scanning campaigns that probe for vulnerable services while minimizing noise by targeting likely open ports.

6.2 Threat Assessment

Focusing on scanner 185.242.226.50, shown in [Table 3](#), it sent over 188,000 packets to nearly 50,000 unique IP addresses while only targeting four ports. This strongly suggests a horizontal scanning strategy, where a single service (e.g., SSH on port 22) is probed across a large address space. The scanner’s behavior is highly consistent with that of a mass scanner or botnet-controlled agent performing service enumeration. Its origin from a cloud-hosting provider in the Netherlands further supports this. Such scanners pose risks by identifying open services for later exploitation, brute-force attacks, or malware delivery.